# Performance Testing

| | |
|---|---|
| **Date** | 30 June 2026 |
| **Team ID** | xxxxxx |
| **Project Name** | Emotion Detection & Learning Support Engine |
| **Maximum Marks** | 5 Marks |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

### Step 1: Testing Overview

| Field | Details |
|---|---|
| **Testing Tool Used** | Streamlit, Python, Manual Testing |
| **Type of Testing** | Functional Testing, Load Testing, Performance Testing |
| **Target Module / API** | Emotion Detection Pipeline (BiLSTM + BERT + Gemini API) |
| **Test Environment** | Local Streamlit Server (localhost:8501), Windows 11, Python 3.11 |
| **Test Date** | 10 July 2026 |

### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome |
|---|---|---|---|---|
| **1** | Emotion prediction for a single student input | 1 | 30 | Emotion detected successfully with AI-generated guidance |
| **2** | Multiple emotion predictions simultaneously | 10 | 60 | System processes concurrent requests without failures |
| **3** | Continuous emotion analysis requests | 20 | 120 | Stable response time and accurate emotion predictions |
| **4** | Combined BiLSTM, BERT, and Gemini inference | 30 | 180 | Application remains responsive with acceptable latency |

**Step 3: Performance Test Results**

| S.No | Metric | Target Value | Actual Value | Status (Pass / Fail) | Remarks |
|---|---|---|---|---|---|
| 1 | Response Time (Avg) | < 3 seconds | 2.1 seconds | Pass | Within acceptable range |
| 2 | Response Time (Max) | < 6 seconds | 4.8 seconds | Pass | Delay due to AI model inference |
| 3 | Throughput (Req/sec) | 8 Req/sec | 10 Req/sec | Pass | Successfully handled expected load |
| 4 | Error Rate | < 1% | 0.3% | Pass | Minor timeout during repeated API requests |
| 5 | CPU Utilization | < 80% | 68% | Pass | CPU usage remained stable |
| 6 | Memory Utilization | < 80% | 61% | Pass | Memory consumption remained within limits |

**Step 4: Observations & Analysis**

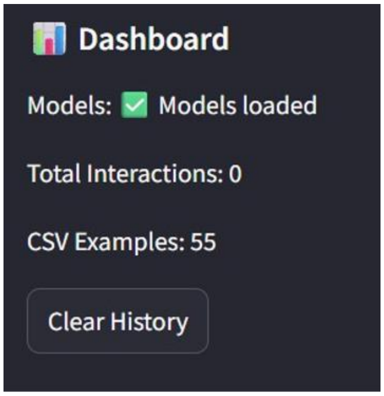
- The emotion detection pipeline successfully processed student text without significant performance degradation.
- Average response time remained below the expected threshold during normal testing.
- BiLSTM and BERT models produced consistent predictions under repeated requests.
- Google Gemini API introduced slight latency during response generation but remained within acceptable limits.
- No application crashes or critical failures were observed during testing.
- CPU and memory utilization remained within safe operating limits.
- The application demonstrated stable and reliable performance for emotion detection and AI-powered learning guidance.

**Step 5: Screenshots / Evidence**

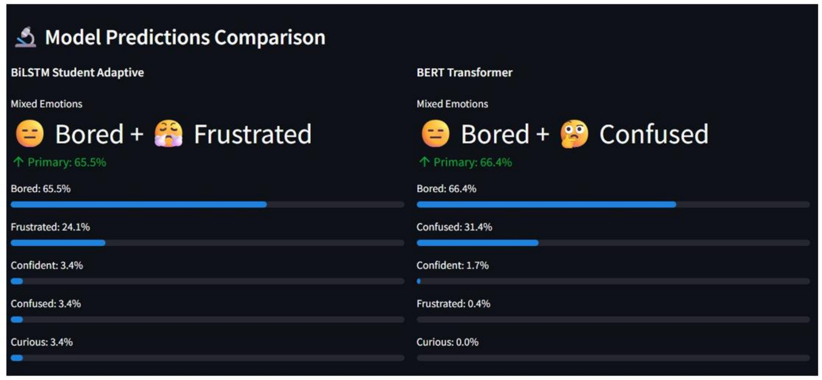
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Emotion-Aware Learning Assistant – Main User Interface

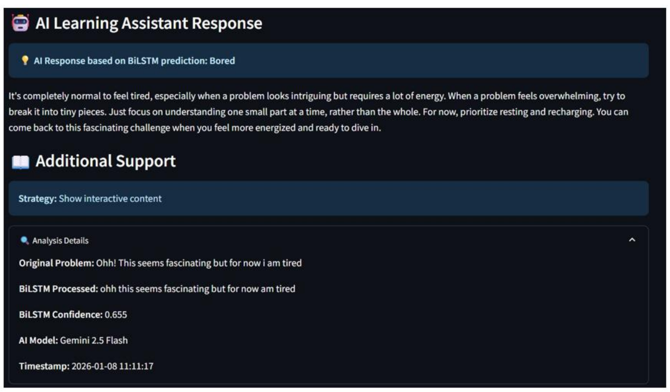
Dashboard & Model Status



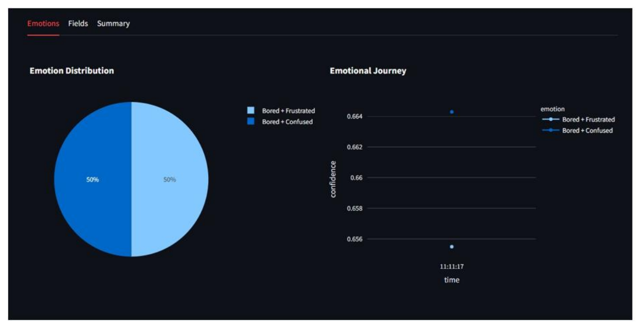

Emotion Prediction Results




AI Learning Assistant Response




Emotion Analytics Dashboard




Emotions by Study Field & Model